HP 200 Operations Manual

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Fictional equipment and specifications created solely for this hackathon. These documents are a closed demonstration dataset and must not be used to operate or maintain real equipment.

Equipment identity and purpose

The fictional HP-200 is an enclosed hydraulic forming press in the demonstration factory. It shapes small metal blanks using a hydraulic ram. Its reference operating fluid is H46 hydraulic oil. Product quality classification and defect causes are not covered by this manual.

Operating reference values

Parameter	HP-200 reference
Normal hydraulic pressure	120 to 140 bar during the forming stage
Oil temperature ready range	35 to 55 degrees C
Oil temperature high warning	Above 60 degrees C
Oil temperature high stop	65 degrees C or higher
Reservoir level indication	Between MIN and MAX sight-glass marks
Pressure-low trigger	Below 110 bar for more than 5 seconds during forming

Machine states

OFF means the demonstration machine is not running. READY requires the guard-closed indicator, oil level within the marked range, and oil temperature in the ready range. FORMING is the stage in which the pressure range applies. HOLD means a fault has interrupted the cycle; a recorded review is required before resumption.

A ready-state reading does not establish the pressure reached during forming. A pressure reading alone does not identify a failed component. Use alarm context and the maintenance reference when answering diagnostic questions.

Document map

Page 2 describes the operator workflow and alarms. Page 3 describes fault reporting and limits of available information. For inspection intervals and maintenance checks, use HP-MNT-01, revision 1.0. It complements this manual and is not a replacement revision.

Operator Workflow and Alarm Reference

Pre-cycle review

In the demonstration workflow, the operator checks the machine ID, reads the displayed state, confirms the guard-closed indicator, checks the reservoir level against MIN and MAX, and records the oil temperature. If a required indication is abnormal, the cycle remains on HOLD and the issue is reported. A guard interlock must never be bypassed.

Alarm codes

Code	Meaning	Documented response
E17	Low hydraulic pressure	Record pressure and cycle stage; place on HOLD and request the checks in HP-MNT-01 pag
E24	Oil temperature high warning	Above 60 degrees C: hold the cycle and request cooling-system inspection. At 65 degrees C
E31	Guard-open indication	Keep the cycle inhibited. Report an indication that remains open when the guard is closed; ne
E42	Return-filter restriction	Record the indicator state and request the filter inspection in HP-MNT-01 page 2.

Interpreting E17

E17 is generated when pressure stays below 110 bar for more than 5 seconds during forming. The normal forming range is 120 to 140 bar. E17 does not prove that a pump has failed. The maintenance manual lists evidence to collect before selecting a cause.

Temperature response

A reading of 62 degrees C is above the warning threshold but below the high-stop threshold. The warning response still requires a hold and cooling-system inspection. A reading of 65 degrees C meets the high-stop condition. The reference does not provide a guaranteed cooldown duration.

Fault Reporting and Scope of Guidance

Information to record

An HP-200 escalation record includes machine ID M-HP200, timestamp, active alarm code, displayed pressure, oil temperature, cycle stage, reservoir indication, visible leak observations, and whether the issue repeated. Record observations separately from suspected causes.

Repeated pressure loss

If E17 repeats after the documented checks have been reviewed, keep the demonstration machine on HOLD and escalate to authorized maintenance. Include the previous incident reference. Do not raise the pressure setting to suppress the alarm. Detailed low-pressure checks are in HP-MNT-01 page 3.

Who reviews the event

The operator records the symptoms. Authorized maintenance reviews inspection findings. The supervisor records the decision on whether the issue has been resolved. The portal may summarize this workflow but must not issue commands to actual equipment.

Relationship to the maintenance manual

HP-OPS-01 supplies operating ranges, alarm definitions, and reporting requirements. HP-MNT-01 supplies inspection intervals and maintenance observations. When a question asks for both the E17 trigger and the first inspection checks, cite the relevant pages from both documents.

Information not provided

This dataset does not specify the pump replacement price, supplier contacts, tightening torque, wiring diagrams, hydraulic circuit drawings, internal repair procedures, pressure adjustment instructions, cycle-time guarantee, or service contract coverage. Questions requiring those details cannot be answered from these manuals.

No conclusion about a future failure date or remaining useful life is supported by this document. An alarm is a reported condition, not a prediction of when the equipment will fail.